OD-Based Multimodal Replicator Dynamics

1 State Variables and Constraints

Let $\mathcal{Z}$ be a set of zones (origins and destinations), and let K be the number of transport modes. Define the set of OD pairs as

$$\mathcal{P} = \{(o,d) \mid o,d \in \mathcal{Z},\ o \neq d\}, \qquad |\mathcal{P}| = N_{\mathrm{od}}.$$

The state of the system is a matrix

$$\mathbf{x} \in \mathbb{R}^{N_{\mathrm{od}} \times K},$$

where $x_{od,m} \in [0,1]$ denotes the *modal share* of mode m for OD pair od. The simplex constraint requires

$$\sum_{m=1}^{K} x_{od,m} = 1, \qquad \forall\, od \in \mathcal{P}.$$

2 Edge Flow Assignment

Each mode m is associated with a graph $G^{(m)}$. For each pair (od, m), a fixed shortest path (computed at free-flow speed) provides a set of edges $\mathcal{E}_{od,m} \subseteq \mathcal{E}$. Given OD demands $q_{od} > 0$, the flow of mode m on edge e is

$$f_{e,m} = \sum_{od \in \mathcal{P}} q_{od}\, x_{od,m}\, \mathbf{1}[e \in \mathcal{E}_{od,m}]\,.$$

3 Edge Cost Function

Let $\ell_e > 0$ be the length of edge e, $v_{e,m} > 0$ the free-flow speed of mode m on edge e and $\kappa_{e,m} > 0$ its effective capacity. The *density* of mode m' on edge e is

$$\rho_{e,m'} = \frac{f_{e,m'}}{\kappa_{e,m}}.$$

The travel cost (congested travel time) for mode m on edge e is

$$\boxed{c_{e,m} \;=\; \underbrace{\frac{\ell_e}{v_{e,m}}}_{\text{free-flow time}} + \underbrace{\sum_{m'=1}^{K} M_{m,m'}\, \rho_{e,m'}}_{\text{congestion term}},}$$

where $M \in \mathbb{R}^{K \times K}$ is the *modal interaction matrix*. The entry $M_{m,m'}$ captures the extent to which the density of mode m' slows down mode m.

4 OD Path Cost

The total cost of using mode m for OD pair od is the sum of edge costs along the corresponding path:

$$C_{od,m} = \sum_{e \in \mathcal{E}_{od,m}} c_{e,m}.$$

5 Replicator Dynamics

Let

$$\bar{C}_{od}(t) = \sum_{m=1}^{K} x_{od,m}(t)\, C_{od,m}(t)$$

be the *average cost* experienced by travelers of OD pair od at time t. The replicator dynamics read

$$\boxed{\dot{x}_{od,m} \;=\; \rho\, x_{od,m}\big(\bar{C}_{od} - C_{od,m}\big)\,,}$$

where $\rho > 0$ is the replication (adjustment) rate.

Interpretation

- A mode whose cost $C_{od,m}$ is *below* the average $\bar{C}_{od}$ is cheaper; its share *grows*.
- A mode whose cost is *above* the average is more expensive; its share *shrinks*.
- The simplex constraint is preserved: $\frac{d}{dt}\sum_m x_{od,m} = \rho \sum_m x_{od,m}(\bar{C}_{od} - C_{od,m}) = \rho(\bar{C}_{od} - \bar{C}_{od}) = 0$.

6 Equilibrium

A fixed point $\mathbf{x}^*$ satisfies $\dot{x}_{od,m} = 0$ for all (od, m), which holds when, for every OD pair od,

$$x^*_{od,m} > 0 \implies C_{od,m}(\mathbf{x}^*) = \bar{C}_{od}(\mathbf{x}^*).$$

This is precisely the Wardrop user-equilibrium condition: no traveler can unilaterally reduce their cost by switching mode.

7 Summary of Notation

Symbol	Meaning
$x_{od,m}$	Modal share of mode m for OD pair od
$f_{e,m}$	Flow of mode m on edge e
$c_{e,m}$	Travel cost of mode m on edge e
$C_{od,m}$	Path cost of mode m for OD pair od
$\bar{C}_{od}$	Average path cost for OD pair od
$M_{m,m'}$	Interaction coefficient (effect of mode m' on mode m)
ρ	Replication rate
ℓ_e, $\kappa_{e,m}\, v_{e,m}$	Edge length, capacity, free-flow speed